

Figure 4: Services tab

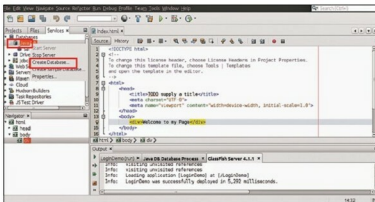


Figure 5: Creating a database

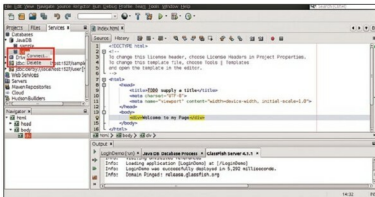


Figure 6: Connecting the database

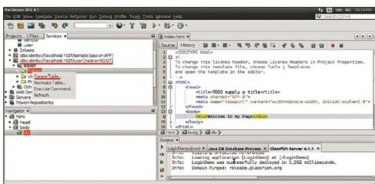


Figure 7: Creating the Table option

Let's try to briefly understand what we are going to make. We will create a database to hold user login information like email, passwords, first names and last names. For model, we will create a Bean name *UserBean* to store email, the first name and last name. We will also create two JSP files, *welcome.jsp* and *error.jsp*, to display the welcome message or error message. Then we will create a controller servlet that will intercept the request and check for login credentials with the database. If the credentials are valid, it will populate the bean and forward the request to *welcome.jsp* to display data; else *error.jsp* comes into the picture and will display an error message.

A step-by-step guide to building a login application using MVC

For this application, we will use the JavaDB database.

To create the database, open the *Services* tab as shown in Figure 4.

Using the *Services* tab, we can configure, add and delete databases, Web servers and many other configurations. To create the new database, expand *Databases* by clicking on the black triangle. This will display the configured databases. For now we can see JavaDB. Right click on JavaDB and select the *Create database* option as shown in Figure 5.

A new window with the title 'Create Java DB' will pop up and ask for the database's name, user name and password.

Remember that the database's name, user name and password will be needed later in querying the database. After filling in the necessary details, click on OK. In the *Services* panel, the user database can be seen in JavaDB. Right click on *User* and click on the *Connect* option as shown in Figure 6.

Once the database is connected, we can see an entry like *jdbc:derby://localhost:1527/user/* in the *Services* panel itself. Expand the entry, which will show ROOT. Expand the ROOT and *Tables* is displayed. Right click on *Tables* and click on *Create table* as shown in Figure 7.

It will open the *Create table* dialogue box. Fill in the table's name. We will use 'login' as the table's name and click on the *Add column* button.

Add column will open a new dialogue box and ask for the column's name, type, size and constraints. Under the column name, enter email, in type: varchar, in constraint: select *Primary Key*.

Clicking OK will display the *Create table* dialogue box. Next, click on *Add column* again;